

Detecting Human-Operated Ransomware with Microsoft Defender XDR

Microsoft Learn field guide for proactive detection and SecOps readiness

Condensed, non-verbatim operational guide

Purpose	Quick-reference guide for detecting new or ongoing human-operated ransomware attacks using Microsoft Defender XDR signals, incidents, and focused SecOps practices.
Primary audience	SecOps analysts, incident responders, threat hunters, Defender XDR administrators, identity and endpoint teams, and security leaders.
Reference source	https://learn.microsoft.com/en-us/defender-xdr/playbook-detecting-ransomware-m365-defender
Generated	2026-07-03
Use note This document is a concise field guide based on Microsoft Learn guidance. Use the linked Microsoft Learn article as the authoritative source for the latest product details, portal behavior, and Defender XDR guidance.	

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1. Detection objective and expected outcome

The goal is to detect pre-ransom and active human-operated ransomware activity early enough for the SOC to disrupt the attack before widespread encryption, exfiltration, or business-impacting downtime occurs.

Key Microsoft Learn framing

The article distinguishes commodity ransomware from human-operated ransomware. Commodity ransomware spreads with phishing or between devices and encrypts files before demanding ransom; human-operated ransomware is planned and coordinated by active cybercriminals using multiple attack methods to infiltrate, find valuable assets or systems, and demand ransom.

2. At-a-glance detection workflow

Step	Phase	Focus
1	Understand the attack type	Separate commodity ransomware from human-operated ransomware so analysts look beyond malware detonation and inspect coordinated adversary behavior.
2	Watch for pre-ransom techniques	Prioritize initial access, reconnaissance, credential theft, lateral movement, and persistence signals, because these can precede encryption.
3	Use Defender XDR incident correlation	Use Microsoft Defender portal incidents to connect small or seemingly unrelated signals that may be part of a larger ransomware campaign.
4	Investigate impacted assets	Review users, devices, apps, identities, endpoints, and cloud-app activity tied to the incident story and evidence set.
5	Disrupt early where possible	If pre-ransom signals are detected early, use smaller-scale mitigations such as isolating infected devices or affected user accounts according to tenant procedures.
6	Prepare the SOC for focus	Create focused ransomware detection practices, hunting routines, and escalation rules so analysts can react quickly to pre-ransom activity.

3. Defender XDR signal coverage called out by Microsoft Learn

The article describes proactive detection with the Microsoft Defender portal as an extended detection and response solution that spans multiple Microsoft security services.

Service	Detection value in this field guide
Microsoft Defender for Endpoint	Endpoint behavior, device investigation, and ransomware-related endpoint signals.
Microsoft Defender for Office 365	Email and collaboration signals that may indicate phishing, payload delivery, or campaign activity.
Microsoft Defender for Identity	Identity and domain-related signals that can reveal credential theft, reconnaissance, or lateral movement.
Microsoft Defender for Cloud Apps	Cloud app activity and app governance signals relevant to suspicious app and SaaS behavior.
Microsoft Entra ID Protection	Identity risk context that can support ransomware campaign detection.
Microsoft Defender for IoT	IoT-related visibility where deployed and relevant to the environment.
Microsoft 365 Business Premium / Defender for Business	Coverage options for smaller organizations using Microsoft Defender capabilities.

4. Focused ransomware detection checklist

Proactive detection readiness

- ☐ Confirm the SOC understands the difference between commodity ransomware and human-operated ransomware.
- ☐ Tune triage processes to look for coordinated attacker behavior instead of only file encryption events.
- ☐ Build escalation criteria for ransomware pre-ransom techniques: initial access, reconnaissance, credential theft, lateral movement, and persistence.
- ☐ Ensure analysts know where to review incidents and evidence in the Microsoft Defender portal.

Pre-ransom signal review

- ☐ Review new incidents that include small but related signals across endpoint, identity, email, cloud app, and user activity.
- ☐ Look for attacker reconnaissance of assets and systems that could be encrypted or extorted.
- ☐ Prioritize signs of credential theft and lateral movement, because these can indicate progression toward ransomware deployment.
- ☐ Treat persistence findings as a high-priority sign that the attacker may be preparing for a longer operation.

Incident correlation and triage

- ☐ Use Defender XDR correlation to determine whether unrelated-looking incidents may belong to a larger campaign.
- ☐ Identify affected users, devices, apps, and services in the incident story and evidence list.
- ☐ Determine whether detected activity is still pre-ransom or has moved into encryption/exfiltration.

Early disruption path

- ☐ If pre-ransom activity is detected early, consider smaller-scale disruptions such as isolating infected devices or affected user accounts according to approved procedures.
- ☐ If detection occurs later during encryption deployment, prepare for more aggressive response actions and expected business disruption.
- ☐ Keep a clear decision log for containment choices, analyst handoffs, and escalations.

SecOps improvement loop

- ☐ Capture detection gaps and missed pre-ransom signals after every ransomware-related investigation.
- ☐ Convert lessons learned into hunting routines, alert tuning, workflow updates, and tabletop scenarios.
- ☐ Cross-train endpoint, identity, email, and cloud-app analysts on the ransomware attack chain.

5. Detailed field guidance

5.1 Prioritize pre-ransom detection

- Human-operated ransomware is often executed by active attackers working through a sequence of actions in real time, so the time taken to detect the attack is critical.
- The Microsoft Learn article says the pre-ransom stage typically includes initial access, reconnaissance, credential theft, lateral movement, and persistence.
- When these techniques are detected early, the likelihood of a severe ransomware attack decreases.

5.2 Connect small signals into an attack story

- The article notes that pre-ransom techniques can initially seem unrelated and often fly under the radar.
- Microsoft Defender XDR can help identify small and seemingly unrelated incidents as possibly part of a larger ransomware campaign.
- Analysts should review incident context across workloads and avoid closing isolated alerts without checking whether they fit an attacker progression.

5.3 Separate early disruption from late-stage response

- If detected during the pre-ransom stage, smaller-scale mitigations such as isolating infected devices or user accounts can be used to disrupt and remediate the attack.
- If detection comes later, such as when encryption malware is being deployed, more aggressive remediation steps may be needed and can cause downtime.
- The article explicitly notes that business operation disruptions are likely when responding to a ransomware attack.

5.4 Use advanced hunting as a proactive supplement

- Use the related Microsoft Learn advanced hunting guidance to look for ransomware-associated activity and signs of compromise.
- Microsoft Learn highlights signs such as stopping processes, turning off services, deleting logs and files, deleting shadow copies, deleting or stopping backups, modifying boot settings, and turning off recovery tools.
- Because some of these activities can be benign, analysts should look for multiple signs on the same devices rather than relying on one isolated event.

5.5 Prepare the SOC for focused ransomware detection

- Define clear triage ownership for endpoint, identity, email, cloud app, and incident correlation signals.
- Create an escalation path from pre-ransom detection to incident response so early evidence is not lost in normal alert queues.
- Review ransomware-related incidents after closure to improve detection logic, hunting queries, and analyst readiness.

6. Ransomware detection evidence matrix

Question	Useful evidence / signal sources
Commodity or human-operated?	Initial alert type, coordinated attacker behavior, multi-workload signals, reconnaissance, credential theft, lateral movement, persistence
Is this pre-ransom activity?	Initial access, reconnaissance, credential theft, lateral movement, persistence, suspicious tools, cross-domain incident correlation

What assets are at risk?	Impacted users, devices, apps, systems, identities, cloud apps, and resources tied to the incident story
Has the attack moved to impact?	Encryption activity, file destruction, exfiltration indicators, ransomware payload deployment, business system disruption
Can early disruption work?	Scope of compromised users/devices, infection spread, attacker control, ability to isolate devices or accounts safely
Will aggressive response be needed?	Encryption deployment, widespread compromise, active exfiltration, business-critical system involvement, containment urgency
What should improve after closure?	Alert tuning, hunting routines, playbooks, training, cross-workload correlation, escalation and executive communication process

7. Detection closure notes template

- ☐ Detection summary: ransomware-related signals observed, whether activity was pre-ransom or active impact stage, and how it was detected.
- ☐ Timeline: first suspicious event, cross-workload correlation, analyst triage, containment decision, and response escalation.
- ☐ Scope: users, devices, identities, apps, cloud apps, systems, and business services involved.
- ☐ Evidence reviewed: Defender XDR incident, alerts, evidence list, advanced hunting results, identity/email/endpoint/cloud app context, and related logs.
- ☐ Verdict: benign, suspicious pre-ransom activity, confirmed human-operated ransomware activity, or confirmed active ransomware impact.
- ☐ Actions completed: device/account isolation, escalation, containment, hunting, responder notification, and documentation.
- ☐ Follow-up recommendations: tuning, new hunting queries, analyst training, focused ransomware detection tabletop, and prevention alignment.

8. Authoritative sources

Primary Microsoft Learn article: [Detecting human-operated ransomware attacks with Microsoft Defender XDR | Microsoft Learn](#)

Related advanced hunting article: [Find Ransomware with Advanced Hunting | Microsoft Learn](#)

Related response playbook: [Responding to ransomware attacks | Microsoft Learn](#)

Related ransomware hub: [Microsoft defense against ransomware, extortion, and intrusion | Microsoft Learn](#)

This field guide does not replace the official articles. Always reference Microsoft Learn for the latest product behavior, portal guidance, and hunting content.